Arithmetic Vacuum Engineering: Prime Channel Muting via Bost-Connes Phase Transitions and Its Implications for Exotic Matter Generation

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We propose a mechanism for controlling the sign of quantum vacuum energy using the arithmetic structure of the Riemann zeta function. The Bost-Connes (BC) quantum statistical mechanical system has partition function $Z(\beta) = \zeta(\beta)$, where each prime p contributes a factor $(1-p^{-\beta})^{-1}$ via the Euler product. We show that projecting out states divisible by a prime p — "muting prime channel p" — modifies the partition function to $\zeta(\beta) \cdot (1 - p^{-\beta})$ and flips the sign of the zeta-regularized vacuum energy: $\zeta_{\neg p}(-3) < 0$ for any prime $p \geq 2$. This sign flip constitutes a violation of the weak energy condition (WEC), which is the principal obstacle to Alcubierre-type warp metrics. We identify three physical mechanisms for implementing prime channel muting (spectral filtering, arithmetic boundary conditions, and phase transition sector selection) and propose a concrete experiment using a superconducting microwave resonator with SQUID-based notch filters to test the arithmetic structure of vacuum energy. The experiment requires existing technology (dilution refrigerator, Nb coplanar waveguide, tunable SQUIDs) at an estimated cost of \$50–100K.

I. INTRODUCTION

The Alcubierre warp metric [1] permits superluminal travel within general relativity, but requires matter that violates the weak energy condition (WEC): a region of negative energy density. The Casimir effect [2] demonstrates that boundary conditions on quantum fields can produce negative vacuum energy, but the magnitude falls short of the Alcubierre requirement by many orders of magnitude [3].

The central observation of this paper is that the quantum vacuum has *arithmetic structure*: its energy is computed via zeta function regularization, and the Riemann zeta function decomposes as an Euler product over primes,

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}. \tag{1}$$

Each prime p contributes an independent "channel" to the vacuum. We show that suppressing a single prime channel — an operation with a precise definition in the Bost-Connes quantum statistical mechanical system [4] — flips the sign of the vacuum energy.

This paper is organized as follows. Section II reviews zeta-regularized vacuum energy and establishes the connection to the Euler product. Section III introduces the Bost-Connes system and defines prime channel muting as a Hilbert space projection. Section IV proves the sign-flip theorem. Section V identifies three physical mechanisms for implementing the muting operation. Section VI proposes a concrete experimental test. Section VII discusses implications for warp drive physics. Section VIII addresses limitations and open problems.

II. ZETA-REGULARIZED VACUUM ENERGY

A. Casimir energy and zeta regularization

The zero-point energy of a quantum field in a cavity with mode spectrum $\{\omega_n\}_{n=1}^{\infty}$ is formally

$$E_{\text{vac}} = \frac{\hbar}{2} \sum_{n=1}^{\infty} \omega_n, \tag{2}$$

which diverges. Zeta function regularization [6, 7] replaces this by

$$E_{\text{vac}}^{\text{reg}} = \frac{\hbar\omega_0}{2}\, \zeta_{\mathcal{H}}(-d), \tag{3}$$

where d is the spatial dimension, ω_0 is the fundamental frequency, and $\zeta_{\mathcal{H}}(s) = \sum_n (\omega_n/\omega_0)^{-s}$ is the spectral zeta function of the Hamiltonian.

For a standard cavity with $\omega_n = n\,\omega_0$, the spectral zeta function reduces to the Riemann zeta function: $\zeta_{\mathcal{H}}(s) = \zeta(s)$. In three spatial dimensions:

$$E_{\text{vac}}^{(3\text{D})} = \frac{\hbar\omega_0}{2}\, \zeta(-3) = \frac{\hbar\omega_0}{2} \cdot \frac{1}{120}. \tag{4}$$

The Casimir energy between parallel plates is the difference between the regularized energy inside and outside the cavity, yielding the well-known result $E/A = -\pi^2\hbar c/(720\, d^3)$ [2].

B. Euler product and prime channels

The Euler product (1) factorizes $\zeta(s)$ into independent contributions from each prime. Defining the *local factor* at prime p:

$$\zeta_p(s) \equiv \frac{1}{1 - p^{-s}}, \tag{5}$$

we have $\zeta(s) = \prod_p \zeta_p(s)$. Each factor $\zeta_p(s)$ encodes the contribution of mode number p (and all its powers) to the full zeta function.

Definition 1 (Prime channel muting). The *p-muted zeta function* is defined by removing the Euler factor at prime p:

$$\zeta_{\neg p}(s) \equiv \zeta(s) \cdot (1 - p^{-s}) = \prod_{q \neq p} \frac{1}{1 - q^{-s}}. \tag{6}$$

Equivalently, $\zeta_{\neg p}(s) = \sum_{n:\, p \nmid n} n^{-s}$ is the Dirichlet series restricted to integers coprime to p. This operation has a geometric interpretation: $\zeta(s)$ is the zeta function of $\mathrm{Spec}(\mathbb{Z})$, while $\zeta_{\neg p}(s)$ is the zeta function of the open subscheme $\mathrm{Spec}(\mathbb{Z}[1/p]) = \mathrm{Spec}(\mathbb{Z}) \backslash \{(p)\}$. Prime muting corresponds to *localization* of the ring of integers.

III. THE BOST-CONNES SYSTEM

A. Definition and partition function

The Bost-Connes (BC) system [4, 5] is a quantum statistical mechanical system defined on the C^*-algebra generated by the Hecke algebra of $\mathbb{Q}/\mathbb{Z}$. Its Hilbert space representation acts on $\ell^2(\mathbb{N}^*)$ with orthonormal basis $\{|n\rangle\}_{n=1}^{\infty}$ and Hamiltonian

$$H\,|n\rangle = \log(n)\,|n\rangle. \tag{7}$$

The partition function at inverse temperature β is

$$Z(\beta) = \mathrm{Tr}(e^{-\beta H}) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta). \tag{8}$$

The BC system exhibits a phase transition at $\beta = 1$ (the pole of ζ). For $\beta > 1$, the symmetry group $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \cong \hat{\mathbb{Z}}^*$ acts on the set of KMS_β states, spontaneously broken.

B. Prime channel muting as Hilbert space projection

Definition 2 (Prime projection). For a prime p, define the projection operator

$$P_{\neg p} = \sum_{\substack{n=1 \\ p \nmid n}}^{\infty} |n\rangle\langle n|, \tag{9}$$

which projects onto the subspace $\mathcal{H}_{\neg p} = \mathrm{span}\{|n\rangle : \gcd(n,p) = 1\}$. The restricted partition function on $\mathcal{H}_{\neg p}$ is

$$\begin{aligned} Z_{\neg p}(\beta) &= \mathrm{Tr}_{\mathcal{H}_{\neg p}}(e^{-\beta H}) = \sum_{\substack{n=1 \\ p \nmid n}}^{\infty} n^{-\beta} \\ &= \zeta(\beta) - p^{-\beta}\,\zeta(\beta) = \zeta(\beta)\,(1 - p^{-\beta}). \end{aligned} \tag{10}$$

This identity is exact and follows from the Euler product decomposition. We emphasize that Eq. (10) is *not* an approximation: it is an algebraic identity valid for all β in the domain of convergence and by analytic continuation to all $s \in \mathbb{C}$.

C. Three physical mechanisms for projection

The projection $P_{\neg p}$ can be physically realized through three distinct mechanisms within the BC framework:

Mechanism A: Spectral filtering. In the BC system, states $|n\rangle$ with $p \mid n$ carry energy contributions of $\log(p)$ from the prime factorization $\log(n) = \sum_q v_q(n) \log(q)$. A notch filter absorbing at frequency $\omega = \log(p)$ selectively removes all p-divisible states.

Mechanism B: Arithmetic boundary conditions. Analogous to the Casimir effect, where conducting plates impose $E_{\mathrm{tangential}} = 0$ boundary conditions that select specific wavelengths, one can impose "arithmetic boundary conditions" that permit only modes n with $\gcd(n,p) = 1$. A periodic structure with period p (a photonic crystal with Bragg spacing $p \cdot a$) opens band gaps at multiples of p, effectively implementing a sieve of Eratosthenes.

Mechanism C: Phase transition sector selection. For $\beta > 1$, the KMS states of the BC system are parameterized by $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \cong \hat{\mathbb{Z}}^* = \prod_p \mathbb{Z}_p^*$. Selecting KMS states invariant under $\mathbb{Z}_p^*$ effectively freezes the p-th component of the Galois action, which is equivalent to projecting out p-divisible states. This is analogous to applying an external magnetic field to select specific spin sectors in a ferromagnet.

IV. THE SIGN-FLIP THEOREM

Theorem 1 (Vacuum energy sign flip)**.** *For any prime $p \geq 2$ and any positive integer n,*

$$\zeta_{\neg p}(1 - 2n) = \zeta(1 - 2n) \cdot (1 - p^{2n-1}). \tag{11}$$

Since $p^{2n-1} \geq 2$ for all primes p and integers $n \geq 1$, the factor $(1 - p^{2n-1})$ is strictly negative. Therefore $\zeta_{\neg p}(1 - 2n)$ and $\zeta(1 - 2n)$ have opposite signs.

Proof. Equation (11) follows directly from Definition II B: $\zeta_{\neg p}(s) = \zeta(s)(1 - p^{-s})$. Evaluating at $s = 1 - 2n$: $1 - p^{-(1-2n)} = 1 - p^{2n-1}$. For $p \geq 2$ and $n \geq 1$: $p^{2n-1} \geq 2^1 = 2 > 1$, so $1 - p^{2n-1} < 0$. □

The physically relevant case is $d = 3$ spatial dimensions, corresponding to $s = -3$ (i.e., $n = 2$):

Corollary 2. *The 3D vacuum energy under p-muting is*

$$\zeta_{\neg p}(-3) = \frac{1}{120}\,(1 - p^3). \tag{12}$$

For $p = 2$: $\zeta_{\neg 2}(-3) = -7/120$. For $p = 3$: $\zeta_{\neg 3}(-3) = -26/120$. Both are strictly negative, corresponding to

TABLE I. Vacuum energy $\zeta_{\neg p}(-3)/\zeta(-3) = (1-p^3)$ for the first several primes. All values are negative, indicating WEC violation for any single-prime muting.

Prime p	$(1-p^3)$	$\zeta_{\neg p}(-3)$
2	-7	$-7/120$
3	-26	$-26/120$
5	-124	$-124/120$
7	-342	$-342/120$
11	-1330	$-1330/120$

negative vacuum energy density and violation of the WEC.

The sign flip is *universal*: it occurs for every prime and grows rapidly with p (Table I). Moreover, the sign oscillates as successive primes are muted: muting an odd number of primes produces negative energy, an even number produces positive energy (relative to the unmuted value). This oscillatory structure is a direct consequence of the multiplicative nature of the Euler product.

For the 1D case ($s = -1$), the sign flip is:

$$\zeta_{\neg p}(-1) = -\frac{1}{12}(1-p) = \frac{p-1}{12} > 0. \tag{13}$$

The unmuted value $\zeta(-1) = -1/12$ is negative; muting any prime makes it positive. This has a direct physical interpretation: the attractive Casimir force becomes repulsive.

V. PHYSICAL MECHANISMS FOR IMPLEMENTATION

The three mechanisms identified in Section III correspond to distinct experimental platforms.

A. Superconducting microwave circuits (Mechanism A)

A coplanar waveguide (CPW) resonator of length L supports modes at frequencies $f_n = nc_{\text{eff}}/(2L)$. SQUID-based notch filters, tuned via external magnetic flux Φ_{ext}, can selectively suppress harmonics at multiples of p.

The dynamical Casimir effect (DCE) has been demonstrated in superconducting circuits [8], proving that vacuum photon creation from boundary modulation is experimentally accessible. Our proposal extends this to *static* boundary conditions that implement arithmetic mode selection.

For $p = 2$: SQUIDs tuned to $2f_0, 4f_0, 6f_0, \ldots$ suppress all even harmonics, leaving only odd modes — the $p = 2$ muted spectrum.

B. Photonic crystals (Mechanism B)

A one-dimensional photonic crystal with period $p \cdot a$ (where a is the fundamental lattice constant) opens Bragg gaps at mode numbers that are multiples of p. Stacking multiple photonic crystals with periods $2a$, $3a$, $5a$ creates an *Euler product crystal*: a sieve of Eratosthenes implemented as a physical structure that transmits only prime-numbered modes.

C. Cold atom optical lattices (Mechanism C)

The BC Hamiltonian $H|n\rangle = \log(n)|n\rangle$ can be quantum-simulated in an optical lattice with site-dependent potential $V_n = V_0 \log(n)$. Adding a secondary lattice of period p with depth Δ shifts the energy of p-divisible sites: $H' = H + \Delta \sum_{p|n} |n\rangle\langle n|$. In the limit $\Delta \to \infty$, p-divisible states are gapped out, realizing $P_{\neg p}$.

The required laser wavelengths are standard: $\lambda_2 = 1064\,\text{nm}$ (Nd:YAG fundamental), $\lambda_3 = 1596\,\text{nm}$, $\lambda_5 = 2660\,\text{nm}$.

VI. PROPOSED EXPERIMENT

A. Circuit design

We propose a $p = 2$ prime filter experiment using a niobium CPW resonator with the following parameters:

- Resonator length: $L = 10\,\text{mm}$
- Effective speed of light: $c_{\text{eff}} = 1.2 \times 10^8\,\text{m/s}$
- Fundamental frequency: $f_0 = \pi c_{\text{eff}}/(2\pi L) \approx 6.0\,\text{GHz}$
- Mode spacing: $\Delta f = f_0 \approx 6.0\,\text{GHz}$

Ten SQUID notch filters are embedded in the resonator, each tuned to an even harmonic ($2f_0$ through $20f_0$). Tuning is achieved via external flux: $\omega_{\text{SQUID}}(\Phi_{\text{ext}}) = \omega_{\text{plasma}}\sqrt{|\cos(\pi\Phi_{\text{ext}}/\Phi_0)|}$. With junction critical current $I_c = 1\,\mu\text{A}$ and capacitance $C_J = 1\,\text{fF}$, the SQUID plasma frequency is $f_{\text{plasma}} \approx 277\,\text{GHz}$, sufficient to cover all target harmonics.

B. Measurement protocol

The experiment compares two configurations at base temperature $T < 20\,\text{mK}$ (dilution refrigerator):

A. **All modes active**: SQUIDs detuned (full spectrum).

B. **Even modes suppressed**: SQUIDs tuned to even harmonics.

For each configuration, we measure: (a) transmission spectrum $S_{21}(f)$, (b) noise power spectral density, (c) photon number $\langle n \rangle$ per mode via dispersive readout.

C. Predicted signal

The zero-point energy of the first $N = 20$ modes is:

$$E_{\text{full}} = \sum_{n=1}^{20} \frac{\hbar\omega_n}{2} \approx 4.2 \times 10^{-22}\,\text{J}, \tag{14}$$

$$E_{\text{odd}} = \sum_{\substack{n=1 \\ n\,\text{odd}}}^{20} \frac{\hbar\omega_n}{2} \approx 2.0 \times 10^{-22}\,\text{J}. \tag{15}$$

The zeta-regularized prediction is that the *ratio* $E_{\text{odd}}/E_{\text{full}}$ approaches $\zeta_{\neg 2}(-1)/\zeta(-1) = (1/12)/(-1/12) = -1$ as the cutoff is removed. The qualitative prediction — sign reversal of the vacuum energy — is cutoff-independent.

D. Error budget

At $T = 15\,\text{mK}$, the thermal occupation of the fundamental mode is $\langle n_{\text{th}} \rangle \approx 5 \times 10^{-9}$ (negligible). SQUID back-action noise contributes $\sim 5 \times 10^{-4}$ photons per mode at $Q = 1000$. On-resonance suppression is 60 dB with notch bandwidth $12\,\text{MHz} \ll f_0 = 6\,\text{GHz}$, ensuring negligible cross-talk to adjacent odd modes.

E. Success criteria

Level 1: Measure $\Delta E \neq 0$ when even harmonics are suppressed (proof that vacuum energy depends on mode arithmetic).

Level 2: Measure $\Delta E/E_{\text{full}}$ consistent with zeta-regularized prediction $(1 - 2^{s+1})$.

Level 3: Repeat with $p = 3$ and $p = 2, 3$ simultaneously; verify multiplicative structure $\Delta E(p_1, p_2) = \Delta E(p_1) \cdot (1 - p_2^{s+1})$.

Level 4: Demonstrate controllable sign-flipping of vacuum energy.

Level 3 success would constitute experimental evidence that the Euler product formula is a *physical law* governing vacuum fluctuations, not merely a mathematical identity.

VII. IMPLICATIONS FOR WARP DRIVE PHYSICS

A. From sign flip to WEC violation

The Alcubierre metric [1]

$$ds^2 = -dt^2 + (dx - v_s f(r_s)\,dt)^2 + dy^2 + dz^2 \tag{16}$$

requires negative energy density in the bubble wall region. The energy density is [3]

$$\rho = -\frac{v_s^2}{32\pi G}\left(\frac{\partial f}{\partial r}\right)^2 \cdot \frac{2}{3} < 0. \tag{17}$$

Theorem 1 shows that prime channel muting produces exactly this: $\zeta_{\neg p}(-3) < 0$, meaning the zeta-regularized vacuum energy density is negative.

The connection between our circuit-scale result and macroscopic spacetime curvature requires bridging the following chain:

1. Arithmetic mode filtering changes vacuum energy (testable).
2. The change matches the zeta-regularized prediction (testable).
3. The Euler product structure of vacuum is a physical law (testable at Level 3).
4. This structure persists at all scales (extrapolation).
5. Negative vacuum energy gravitates (standard GR assumption).
6. Sufficient negative energy can be concentrated (open problem).

Steps 1–3 are experimentally addressable by our proposal. Steps 4–6 remain theoretical challenges.

B. Energy scale and the scaling problem

The energy in our circuit experiment is $|\Delta E| \sim 10^{-22}$ J. The Alcubierre requirement for a 50 m bubble at $v = c$ is $\sim 10^{47}$ J. This gap of $\sim 10^{69}$ orders of magnitude is the *scaling problem*.

We note, however, that Theorem 1 is a statement about *sign*, not magnitude. The sign of the vacuum energy is a topological invariant — it is either positive or negative, independent of the energy scale. Whether this discrete sign flip can be leveraged at macroscopic scales without proportional energy input remains an open and critical question.

Several theoretical mechanisms have been proposed for bridging the gap: coherent amplification ($E \propto N^2$ for N synchronized systems), topological protection of vacuum energy states, and holographic amplification via AdS/CFT-type correspondences. None are yet established.

C. Connection to recent warp drive literature

Lentz [11] demonstrated that soliton solutions with purely positive energy can produce warp-like effects at subluminal velocities. Bobrick and Martire [12] provided a general classification of warp drive spacetimes, showing that subluminal positive-energy solutions exist but superluminal travel requires WEC violation.

Our work complements these results: if superluminal warp requires WEC violation, we provide a specific algebraic mechanism (prime channel muting) that produces it, along with an experimental protocol to test the underlying physics.

VIII. DISCUSSION

A. What the experiment tests

It is important to distinguish what our proposed experiment can and cannot demonstrate:

- **Can test**: Whether suppressing specific harmonics in a microwave resonator changes the vacuum energy in a manner consistent with zeta regularization.
- **Can test**: Whether the Euler product structure manifests as a multiplicative factorization of vacuum energy contributions.
- **Cannot test**: Whether such vacuum energy modifications curve spacetime or produce exotic matter at macroscopic scales.

A positive result at Level 3 would establish that the arithmetic structure of the zeta function is physically realized in the quantum vacuum. This would be a significant result in fundamental physics regardless of its implications for warp drive.

B. Relation to noncommutative geometry

The Bost-Connes system arises naturally in Connes' noncommutative geometry (NCG) program [5, 10]. In the NCG approach to the Standard Model, spacetime is $M^4 \times F$ where F is a finite noncommutative space. The BC system encodes the arithmetic structure of F as a C^*-dynamical system whose KMS states are classified by the absolute Galois group $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$.

The precise sense in which "operating on the BC system modifies spacetime" requires the identification:

> *The internal space F in Connes' $M^4 \times F$ contains, as its arithmetic skeleton, the BC C^*-algebra $C^*(\mathbb{Q}_+^* \backslash \mathbb{A}_f / \hat{\mathbb{Z}}^*)$.*

This identification is established in [5], Theorem 4.1 and Chapter 3. However, the physical implications — that modifications of the BC partition function correspond to modifications of the vacuum state of quantum fields on M^4 — remain a conjecture that our experiment is designed to probe.

C. Limitations

1. The mapping from BC system to physical spacetime relies on the NCG framework, which is not yet experimentally confirmed beyond its successful reproduction of the Standard Model Lagrangian.

2. The scaling problem (Section VII) is open. Our sign-flip theorem is algebraically exact, but whether the sign flip at the regularized level produces macroscopic negative energy density is a separate (and harder) question.

3. The proposed experiment measures vacuum energy in an engineered system (a resonator), not in free space. The connection between resonator vacuum and free-space vacuum requires the Casimir-effect paradigm: that boundary conditions physically modify the vacuum. This paradigm is experimentally established [2, 9] but its extension to "arithmetic boundaries" is new.

IX. CONCLUSION

We have shown that the Bost-Connes quantum statistical mechanical system provides a precise mathematical framework for "prime channel muting" — the suppression of vacuum modes at multiples of a prime p. The resulting vacuum energy $\zeta_{\neg p}(-3)$ is negative for *every* prime $p \geq 2$, violating the weak energy condition.

Three physical mechanisms for implementation have been identified: spectral filtering via SQUID notch filters, arithmetic boundary conditions via photonic crystals, and phase transition sector selection via quantum simulation. A concrete experimental protocol using existing superconducting circuit technology has been proposed, with an estimated cost of \$50–100K and a 12-month timeline.

If the experiment confirms that arithmetic mode selection modifies vacuum energy as predicted by zeta regularization, it would establish that the Euler product decomposition of $\zeta(s)$ is not merely a mathematical identity but a physical law governing the quantum vacuum. This would open the door to *arithmetic vacuum engineering* — the systematic control of vacuum energy through number-theoretic design of boundary conditions — with potential long-term implications for exotic matter generation and warp drive physics.

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[1] M. Alcubierre, "The warp drive: hyper-fast travel within general relativity," Class. Quantum Grav. **11**, L73–L77 (1994).

[2] H. B. G. Casimir, "On the attraction between two perfectly conducting plates," Proc. K. Ned. Akad. Wet. **51**, 793–795 (1948).

[3] M. J. Pfenning and L. H. Ford, "The unphysical nature of 'warp drive'," Class. Quantum Grav. **14**, 1743–1751 (1997).

[4] J.-B. Bost and A. Connes, "Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory," Selecta Math. (N.S.) **1**, 411–457 (1995).

[5] A. Connes and M. Marcolli, *Noncommutative Geometry, Quantum Fields and Motives*, AMS Colloquium Publications, vol. 55 (2008); see also A. Connes and M. Marcolli, "From physics to number theory via noncommutative geometry," in *Frontiers in Number Theory, Physics, and Geometry I*, Springer, pp. 269–347 (2006).

[6] S. W. Hawking, "Zeta function regularization of path integrals in curved spacetime," Commun. Math. Phys. **55**, 133–148 (1977).

[7] E. Elizalde, *Ten Physical Applications of Spectral Zeta Functions*, Springer Lecture Notes in Physics **855** (2012, 2nd ed.).

[8] C. M. Wilson, G. Johansson, A. Pourkabirian, M. Simoen, J. R. Johansson, T. Duty, F. Nori, and P. Delsing, "Observation of the dynamical Casimir effect in a superconducting circuit," Nature **479**, 376–379 (2011).

[9] S. K. Lamoreaux, "Demonstration of the Casimir force in the 0.6 to 6 μm range," Phys. Rev. Lett. **78**, 5–8 (1997).

[10] A. Connes, *Noncommutative Geometry*, Academic Press (1994).

[11] E. W. Lentz, "Breaking the warp barrier: hyper-fast solitons in Einstein-Maxwell-plasma theory," Class. Quantum Grav. **38**, 075015 (2021).

[12] A. Bobrick and G. Martire, "Introducing physical warp drives," Class. Quantum Grav. **38**, 105009 (2021).

[13] T. H. Boyer, "Quantum electromagnetic zero-point energy of a conducting spherical shell and the Casimir model for a charged particle," Phys. Rev. **174**, 1764–1774 (1968).

[14] M. Bordag, G. L. Klimchitskaya, U. Mohideen, and V. M. Mostepanenko, *Advances in the Casimir Effect*, Oxford University Press (2009).

[15] A. Kitaev, "Periodic table for topological insulators and superconductors," AIP Conf. Proc. **1134**, 22–30 (2009).

[16] D. Quillen, "Higher algebraic K-theory: I," in *Algebraic K-theory I*, Springer LNM **341**, pp. 85–147 (1973).